

42577 - Introduction to Business Analytics

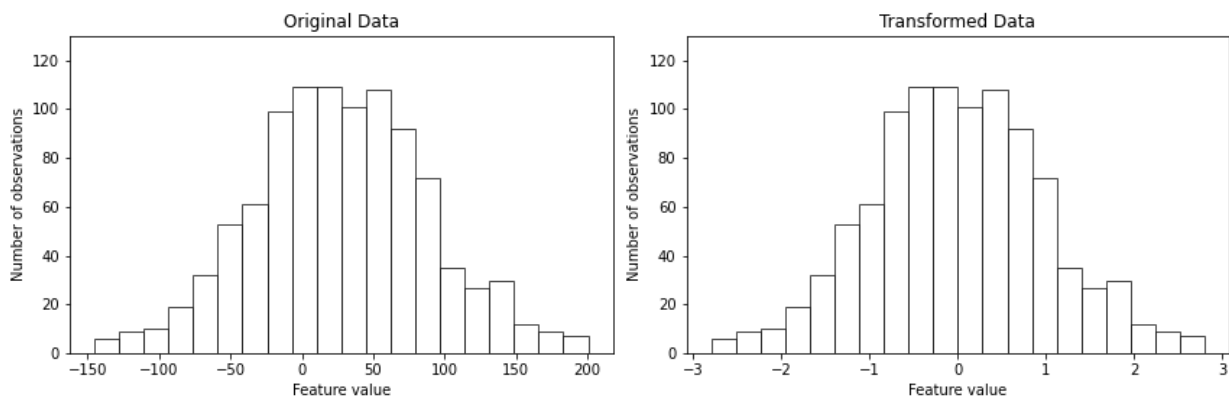
Quiz – Example questions

Name and Surname:

Student Number:

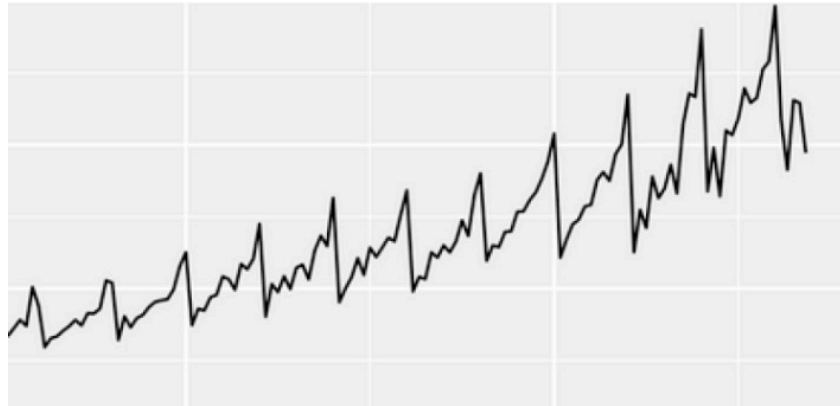
Instructions: The quiz consists of XX multiple-choice questions. The multiple-choice questions may have multiple options that are correct. The questions that have multiple correct options are indicated in the text. Please circle all the correct options. You receive 1 point for a correct answer and 0 points for a wrong answer (in case multiple options are correct you need to select all the correct options to get 1 point). There is no negative marking. Please indicate your name, surname, and student number.

1. Look at the following figure. The image on the left, shows the initial dataset (left) show the distribution of the original data, with the x-axis showing the numerical value of the feature, while the y-axis depicts the number of observations associated to that value. In order to combine this data with others, the modeller has transformed the data. The distribution for the transformed data is on the right. Can you say whether the data have been standardized, normalized, or if a different transformation was used?



- A. The plot on the right represents the standardized version of the data on the left.
- B. The plot on the right represents the normalized version of the data on the left.
- C. The plot on the right represents a transformed version of the original data set, which is not normalization nor standardization.
2. In which of the following scenarios would Multidimensional Scaling (MDS) be more appropriate than Principal Component Analysis (PCA)?
- A. When you need to reduce dimensionality in a dataset with linear relationships and want to maximize the variance explained by a small number of components.
- B. When the dataset contains non-linear relationships and you want to preserve the original pairwise distances between points in the lower-dimensional space.
- C. When the primary goal is to cluster the data based on categorical features rather than numerical distances or variances.
- D. When the dataset is highly correlated and the goal is to retain only the features with the highest variance.

3. You are analyzing a sample of time-series data which is shown in the figure below. In order to understand the nature of the time-series patterns in your data, which of the following approaches would you use?



- A. Additive decomposition
 B. **Multiplicative decomposition**
 C. Either the additive or the multiplicative decomposition would work
 D. None of the above
4. In the context of machine learning, what is the bias-variance trade-off?
- A. **It refers to the balance between using a simple model with high bias and a complex model with high variance.**
 B. It represents the trade-off between the accuracy of training data and the reliability of test data.
 C. It describes the trade-off between the number of training examples and the number of features in a dataset.
 D. It refers to the trade-off between precision and recall in classification tasks.
5. Assume that you generate a synthetic data set of 10000 observations with a single feature X. You then generate a response variable Y that is a known 5th-degree polynomial function of X (true model) plus a random error term (normally distributed with zero mean and variance 3.5). You split the data set into a training set (6000 observations) and a test set (4000 observations). You now train three models using your training set: M1 is a linear model, M2 is a fourth-degree polynomial function and M3 is a degree 12 polynomial. You observe the following training and test MSE for the three models from your data set.

Model	Training MSE	Test MSE
M1 (linear)	30	32
M2 (degree 4 polynomial)	4	5
M3 (degree 12 polynomial)	3	18

Which of the following statements are true? (**more than one option can be correct**)

- A) M1 has high variance and low bias
 B. **The test MSE of all three models estimated will always be larger than 3.5**
 C. **M1 has low variance but high bias**
 D) M3 has low variance and low bias
6. Explain the concept of dropout in neural networks and its role in improving model generalization. Additionally, discuss potential drawbacks or limitations of using dropout in deep learning models